Ideal Chains Do Not Measure Bratteli Path Depth: A Disproof of Conjecture 00000009783

GitHub: gaochengzhecpu

Abstract

The AF algebra $\mathbb{C}$ has exactly two closed ideals and hence a unique maximal ideal chain, of length one. Its standard constant one-dimensional inductive system has a Bratteli diagram with one vertex at each level and one edge between consecutive levels. This diagram has paths of every finite length. The claimed equality between ideal chain length and the supremum of path depths therefore fails. The counterexample remains valid if chain length counts elements instead of strict inclusions. Lean verifies the actual complex algebra, its finite-dimensional approximation and direct-limit property, its closed ideals, and genuine paths in the associated diagram.

Reading of the source

Both language versions assert that the length of maximal ideal chains in an AF algebra equals the supremum of path depths of its Bratteli diagram. We use the standard meanings: closed two-sided ideals ordered by inclusion, and depth of a finite directed path measured by its number of edges. Counting vertices changes path length by one and does not change whether depths are unbounded. Counting the elements of an ideal chain instead of its strict inclusions also leaves the counterexample intact.

An AF algebra is a norm closure of an increasing union of finite-dimensional C^*-subalgebras, equivalently an inductive limit of finite-dimensional C^*-algebras. Finite-dimensional algebras are included: their constant identity systems are allowed. A Bratteli diagram has a vertex for each simple matrix summand at each stage, and the number of edges between successive-stage vertices is the partial embedding multiplicity. These conventions are stated, for example, in Blackadar's author-hosted text, sections I.3.2.1–2 and I.3.2.6–8 [1].

We refute the chain-depth clause. We do not need to choose between conventions on hereditary subsets or their saturation, nor make any assertion about the separate dimension-vector stratification clause. Changing those conventions cannot change the two ideals of the algebra or the unbounded path lengths of the specified diagram.

Theorem 1. *There is an AF algebra with a Bratteli diagram whose finite path depths are unbounded, but whose unique maximal closed-ideal chain has two elements and one strict inclusion.*

Proof. Take $A = \mathbb{C}$, with its usual norm, multiplication and complex conjugation. It is a unital complex C^*-algebra. Put

$$A_n = \mathbb{C} = M_1(\mathbb{C}), \qquad \varphi_n = \mathrm{id}_{\mathbb{C}} \quad (n \geq 0).$$

Internally all stages are the entire closed star subalgebra of A. They are one-dimensional, form an increasing sequence, and their union is exactly A. Hence A is AF.

For clarity, this system really has A as its inductive limit. The canonical maps from all stages into A are identities. For any C^*-algebra B, a compatible family of star algebra maps $f_n : \mathbb{C} \to B$ satisfies $f_{n+1} = f_n$. It therefore factors through the unique map $f_0 : A \to B$. These maps are continuous: they are complex-linear maps with finite-dimensional domain. The same

argument also applies to nonunital star algebra maps and nonunital targets. This is the required direct-limit universal property.

The explicit identification $\mathbb{C} \to M_1(\mathbb{C})$ sends z to the matrix $[z]$ and respects the star operation. Under this identification, each connecting map is the identity on $M_1(\mathbb{C})$. It contains exactly one copy of the source matrix block, so its multiplicity is one. There is consequently one vertex v_n at each level and exactly one edge $v_n \to v_{n+1}$:

$$v_0 \longrightarrow v_1 \longrightarrow v_2 \longrightarrow \cdots .$$

For every $N \geq 0$, the path from v_0 through $v_1, \ldots, v_N$ has exactly N edges. Thus the supremum of finite path depths is $+\infty$.

An ideal I of $\mathbb{C}$ containing a nonzero element z contains $z^{-1}z = 1$, so it is all of $\mathbb{C}$. Therefore the only ideals are $\{0\}$ and $\mathbb{C}$, and both are closed in the norm topology. The complete closed-ideal lattice is the two-element chain

$$\{0\} \subsetneq \mathbb{C}.$$

It is the unique maximal chain, since every subset can be extended to this whole two-element chain. Its length is one by strict inclusions, or two by elements. Neither finite value equals $+\infty$. This disproves the asserted equality under either chain-length convention. □

Formal verification and scope

The Lean file uses Mathlib's actual complex numbers with the `CommCStarAlgebra` instance. Its predicate `ApproximatelyFinite` requires a monotone sequence of actual `StarSubalgebra` objects, each closed and finite-dimensional, whose union is dense. The counterexample satisfies it with every stage equal to the top subalgebra. The actual stage types are identified with $M_1(\mathbb{C})$ by star algebra equivalences.

The connecting maps are actual `StarAlgHom` identities. The theorem

`constant_system_universal`

proves existence and uniqueness of the factorization for every unital C^*-algebra target and compatible unital cocone; continuity of each map out of a stage is proved separately. A second theorem proves the universal property, including continuity, for nonunital C^*-algebra targets and nonunital star algebra maps as well. The internal stage inclusions and external identity system are checked to agree. Conjugating the connecting map by the matrix equivalence gives the actual identity matrix-block embedding. In this one-dimensional case the multiplicity is the complex dimension of its range, which is proved to be one. No general formula identifying the dimension of an arbitrary matrix-algebra image with a partial embedding multiplicity is asserted.

The diagram is a genuine Mathlib `Quiver` on $\mathbb{N}$: its arrow type from n to m is a finite type with the computed multiplicity when $m = n + 1$, and is empty otherwise. The recursively constructed `Quiver.Path` from 0 to N has length N. The supremum over lengths of all actual paths is taken in $\mathbb{N} \cup \{+\infty\}$ and is proved to equal $+\infty$.

`ClosedIdeal` consists of actual Mathlib ideals with a proof that their carriers are topologically closed. Since the algebra is commutative, these are its closed two-sided ideals. Lean proves that each is the zero or whole ideal, that these are distinct, and that there are exactly two. It also proves that every maximal chain, using Mathlib's `IsMaxChain`, is the whole ideal set.

The maximum strict-inclusion length is encoded by the supremum of n for which there is a strictly increasing map from $\{0, \ldots, n\}$ into the closed-ideal order. Lean proves that this supremum is one, and that the corresponding element-count supremum is two. These are computations of chains in the actual ideal order, not stipulated numerical invariants.

The proposition `OneBlockDepthClause` quantifies over all commutative complex C^*-algebras that are AF and star-algebra isomorphic to $\mathbb{C}$. It demands that their ideal-chain length equal the

depth of the constant one-block diagram. Each such algebra admits the same identity-system presentation. This is a necessary restriction of the source's claim about all AF algebras; its negation is the main theorem

`conjecture_9783_false`.

The alternate element-count convention is separately refuted.

No general AF classification theorem, hereditary-subset correspondence, or dimension-vector stratification theorem is claimed as formalized. The exact counterexample, the direct-limit property, the finite matrix-block identification, the ideals and the paths are all verified. The Lean file uses no `sorry`, `admit`, `native_decide`, or custom axiom. Its axiom reports contain only `propext`, `Classical.choice` and `Quot.sound`.

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References

[1] B. Blackadar, *Classification of C^*-Algebras*, incomplete preliminary edition, June 4, 2025, sections I.3.2.1–2 (AF algebras) and I.3.2.6–8 (multiplicities and diagrams). Author's public text.